

ANFENG XU

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EDUCATION

University of Southern California (USC) Ph.D. in Electrical Engineering, Minor in Computer Science	Expected Dec 2026 GPA: 3.96/4.0
University of Southern California (USC) M.S. in Electrical Engineering	May 2024 GPA: 4.0/4.0
University of California, San Diego (UCSD) B.S. in Electrical Engineering, Minor in Mathematics Summa Cum Laude (Top 2% GPA)	June 2021 GPA: 3.97/4.0

RESEARCH EXPERIENCE

Speech Language Models, Speech Processing, Large Language Models, Multimodal Modeling, ASR, Healthcare AI.

WORK EXPERIENCE

Meta Superintelligence Labs <i>Ph.D. Research Scientist Intern, Advisor: Zhong Meng</i>	May 2026 - Aug 2026 <i>Menlo Park</i>
Conducted research on reinforcement learning, including GRPO, for speech language models.	
Meta Superintelligence Labs <i>Ph.D. Research Scientist Intern, Advisors: Yashesh Gaur, Naoyuki Kanda</i>	May 2025 - Aug 2025 <i>Seattle</i>
Developed speech naturalness prediction models for conversational speech generative models (e.g., LLaMA Full-Duplex). Built a large-scale synthetic data augmentation pipeline using internal speech generative models.	
Meta Reality Labs <i>Ph.D. Research Scientist Intern, Advisor: Biqiao Zhang</i>	May 2024 - Aug 2024 <i>Seattle</i>
Conducted research to enhance the Keyword Spotting (KWS) models for wearable devices.	
Signal Analysis and Interpretation Laboratory (SAIL), USC <i>Ph.D. Research Assistant, Advisor: Shrikanth Narayanan</i>	August 2021 - Present <i>Los Angeles</i>
Conduct research projects with the main focus on speech & language processing and multi-modality. Lead research projects, mentor MS and junior Ph.D. students, and manage the IEMOCAP/CreativeIT data distribution.	

RESEARCH PROJECTS

Automated Speech Behavior Analysis and Scoring (USC - Simons Foundation)	Mar 2026 – Present
Developing automated methods to quantify speech behaviors in child and older adult populations with speech foundation models, LLMs, and interpretable features.	
Low-Resource ASR and Speaker Diarization (USC - Simons Foundation)	Oct 2024 - Mar 2026
Benchmarked speech foundation models for speaker diarization across the lifespan. Developed LLM error correction methods for ASR for child-adult speech. Developed an efficient ASR model with integrated speaker and timestamp prediction using serialized output training, jointly trained for speaker diarization.	
Speaker Diarization for Child-Adult Interactions (USC - BU - Apple)	Jan 2022 - Oct 2024
Proposed annotation framework and modeling pipeline to predict language capabilities of children directly from audio. Developed an audio-visual child-adult speaker classification method. Built a child-adult speaker diarization model with Speech Foundation Models. Built simulated conversations from AudioSet to further enhance child-adult speaker diarization.	

SELECTED PUBLICATIONS

- T Feng, J Lee, **A Xu**, et al. “Vox-Profile: A Speech Foundation Model Benchmark for Characterizing Diverse Speaker Traits and Speech Properties,” accepted to Data-centric Machine Learning Research (**DMLR**), 2026.
- T Feng, **A Xu**, et al. “ChildVox: A Speech, Audio, and Large Audio-Language Model Benchmark in Understanding and Characterizing Sound across Childhood,” accepted to **EMNLP Main**, 2026.
- M Nicholas, K Zareno, K Avramidis, **A Xu**, and S Narayanan. “Speech Entrainment in Multi-Party Conversations with a Digital Agent,” **Interspeech**, 2026.
- A Xu**, T Feng, S Bishop, C Lord, S Narayanan. “End-to-End Joint ASR and Speaker Role Diarization with Child-Adult Interactions,” IEEE/ACM Transactions on Audio, Speech and Language Processing (**TASLP**), 2026.
- T Feng, K Huang, **A Xu**, et al. “Voxlect: A Speech Foundation Model Benchmark for Modeling Dialects and Regional Languages Around the Globe,” **KDD**, 2026.
- A Xu***, T Feng*, SH Kim, S Bishop, C Lord, S Narayanan. “Large Language Models based ASR Error Correction for Child Conversations,” **Interspeech**, 2025
- T Feng, **A Xu**, X Shi, S Bishop, S Narayanan. “Egocentric speaker classification in child-adult dyadic interactions: From sensing to computational modeling,” **Interspeech**, 2025
- Z Shi, X Shi, **A Xu**, T Feng, H Srivastava, S Narayanan, M Mataric. “Examining Test-Time Adaptation for Personalized Child Speech Recognition,” **Interspeech**, 2025
- A Xu**, T Feng, H Tager-Flusberg, C Lord, S Narayanan. “Data Efficient Child-Adult Speaker Diarization with Simulated Conversations,” **ICASSP**, 2025.
- A Xu**, B Zhang, S Kong, Y Huang, Z Yang, S Srivastava, M Sun. “Effective Integration of KAN for Keyword Spotting,” **ICASSP**, 2025.
- T Feng, **A Xu**, et al. “Can Generic LLMs Help Analyze Child-Adult Interactions Involving Children with Autism in Clinical Observation?” **NeurIPS Workshop** (GenAI for Health), 2024.
- A Xu**, K Huang, T Feng, L Shen, H Tager-Feng, S Narayanan. “Exploring Speech Foundation Models for Speaker Diarization in Child-Adult Dyadic Interactions,” **Interspeech**, 2024.
- A Xu**, K Huang, T Feng, H Tager-Feng, S Narayanan. “Audio-visual child-adult speaker classification in dyadic interactions,” **ICASSP**, 2024.
- D Bose, R Hebber, T Feng, K Somandepalli, **A Xu**, S Narayanan. “MM-AU:Towards Multimodal understanding of advertisement videos,” **ACM Multimedia**, 2023.
- A Xu**, R Hebbar*, R Lahiri*, T Feng*, L Butler, L Shen, H Tager-Flusberg, and S Narayanan. “Understanding Spoken Language Development of Children with ASD Using Pre-trained Speech Embeddings,” **Interspeech**, 2023.
- (* equal contribution)

SELECTED PREPRINTS / MANUSCRIPTS UNDER REVIEW

- A Xu**, Y Gaur, N Kanda, Z Ouyang, K Žmolíková, D Raj, S Merello, A Sun, O Kalinli. “Conversational Speech Naturalness Predictor,” arXiv:2603.01467.
- A Xu**, et al. “Speaker Role and Language Diarization for Analyzing Multilingual Interviews for Language Proficiency of Older Adults,” arXiv:2608.09032, under review for Computer Speech & Language.
- T Feng, **A Xu**, J Lee, S Narayanan. “VoxCog: Towards End-to-End Multilingual Cognitive Impairment Classification through Dialectal Knowledge,” arXiv:2601.07999, under review for Computer Speech & Language.

AWARDS

- MHI Ph.D. Scholar Finalist (USC, 2026)
- ECE Ph.D. Screening Exam (USC, 2022): Ranked 1st for the Signals and Systems track.
- Annenberg Fellowship (USC, 2021): Top-off fellowship selected among incoming Ph.D. students.
- Summa Cum Laude (UCSD, 2021): Highest Distinction for GPA within the top 2% of the graduating class.

LANGUAGES

English (fluent), Japanese (native), Mandarin Chinese (conversational).

TEACHING ASSISTANT

EE641: Deep Learning Systems – Ph.D. level (USC, Fall 2026)

SERVICES

Reviewer - ICASSP, Interspeech, ACM-MM, IEEE Open Journal of Signal Processing.

Organizer - IEEE ASRU 2025 Satellite Workshop-AI for Children's Speech and Language.